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HELIUM SCATTERING AS A PROBE OF THE CLEAN AND ADSORBATE COVERED Pt(111) SURFACE

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Thermal energy (16–165 meV) helium nozzle beam scattering has been used as a probe of the physical structure of the nominal Pt(111) surface and for the consequential effects of the adsorption of gases including CO, H_2, O_2 and C_2H_2. By careful crystal alignment and polishing techniques a Pt(111) surface with terrace widths greater than 3000 Å was achieved as judged by the helium scattering. The profile of the scattered peak was immeasurably affected by the presence of various adsorbates while the intensity decreases monotonically with coverage. The sensitivity of the scattered intensity to adsorbates is, in general, a function of the energy and angle of incidence of the helium beam. The scattering cross-section of the adsorbate can be, for some cases, much larger than its Van der Waals diameter and apparently increases with the degree of perfection of the surface.

1. Introduction

The scattering of thermal energy beam of atoms from surfaces can be used as a sensitive probe for the surface per se as well as for the investigation of adsorbates, as shown recently, for example, for the H_2/Pt(997) system [1]. The purpose of this study is to determine whether this method can be applied more generally. This goal is best achieved when surface structural effects are minimized so that the consequential effects of adsorbates are most clearly revealed. Therefore, we have chosen to study scattering from the Pt(111) surface because, being a densely packed metal surface it is, ideally, nearly structure-free for thermal energy helium scattering.

2. Experimental

The helium scattering experiments reported here were performed in an UHV supersonic nozzle beam system that has been described previously [1]. For the present study, the detector was operated at the maximum angular

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resolution of 0.15° which, coupled with the direct beam divergence of 0.20°, gives a net measured angular width of 0.25° (FWHM).

The platinum single crystal was cut from a 12 mm diameter rod obtained from Metals Research, England and polished mechanically. Alignment of the polished surface relative to the (111) plane was accomplished using Laue X-ray diffraction. After alignment and polishing, the crystal was cleaned and annealed in the UHV molecular beam apparatus by a combination of Ar^+ sputtering and O_2 and H_2 exposures coupled with heating to temperatures as high as 1500 K. After these treatments,the crystal was judged "clean" by AES and "single crystal" by LEED. However, the helium atom scattering revealed structural features, as discussed below, and the crystal was again oriented and polished, this time with much better results. The second time even more care was taken in orienting ($\pm 0.1°$) and polishing the crystal [2].

3. Results

Figs. 1a–1d show results for helium scattering from the Pt(111) surface after the first preparation. Figs. 1e and 1f give results after re-polishing the surface. The figures on the left side present the measured peak height of the specularly scattered beam relative to the direct beam as a function of scattering angle. The figures on the right show normalized direct and scattered beam profiles for the conditions corresponding to the figures on the left.

Figs. 1a–1d show clear signs of surface structure, the main features of which can be understood in terms of a particle-wave interacting with two or more parallel planes. The theory of interactions with structured surfaces has been previously developed in conjunction with LEED studies [3] and for randomly stepped surfaces using atom scattering by Lapujoulade [4] and will not be reviewed in any detail here. The peak height oscillations in figs. 1a and 1c appear to result from the interference of scattering contributions from some type of step structure (e.g., random or monotonic).

The path difference for scattering from two adjacent layers is given by

$$\Delta l = n\lambda = 2h \cos \vartheta_i, \tag{1}$$

where λ is the de Broglie wave length of the helium atom, h is the interlayer spacing, and n has integer values for constructive interference (maxima) and half-integer values for destructive interference (minima). The positions of the arrows in figs. 1a and 1c correspond to the expected positions of the maxima and minima for h equal to the (111) interlayer spacing of 2.27 Å.

The amplitude of the peak height oscillations is related to the surface step density. Assuming a random distribution of terrace widths, an estimate can be made following the approach of Lapujoulade [4] which gives an "average" terrace width of roughly 300 Å. Alternatively, one can assume uniform, monotonic steps and calculate a similar value which would imply a misalignment of

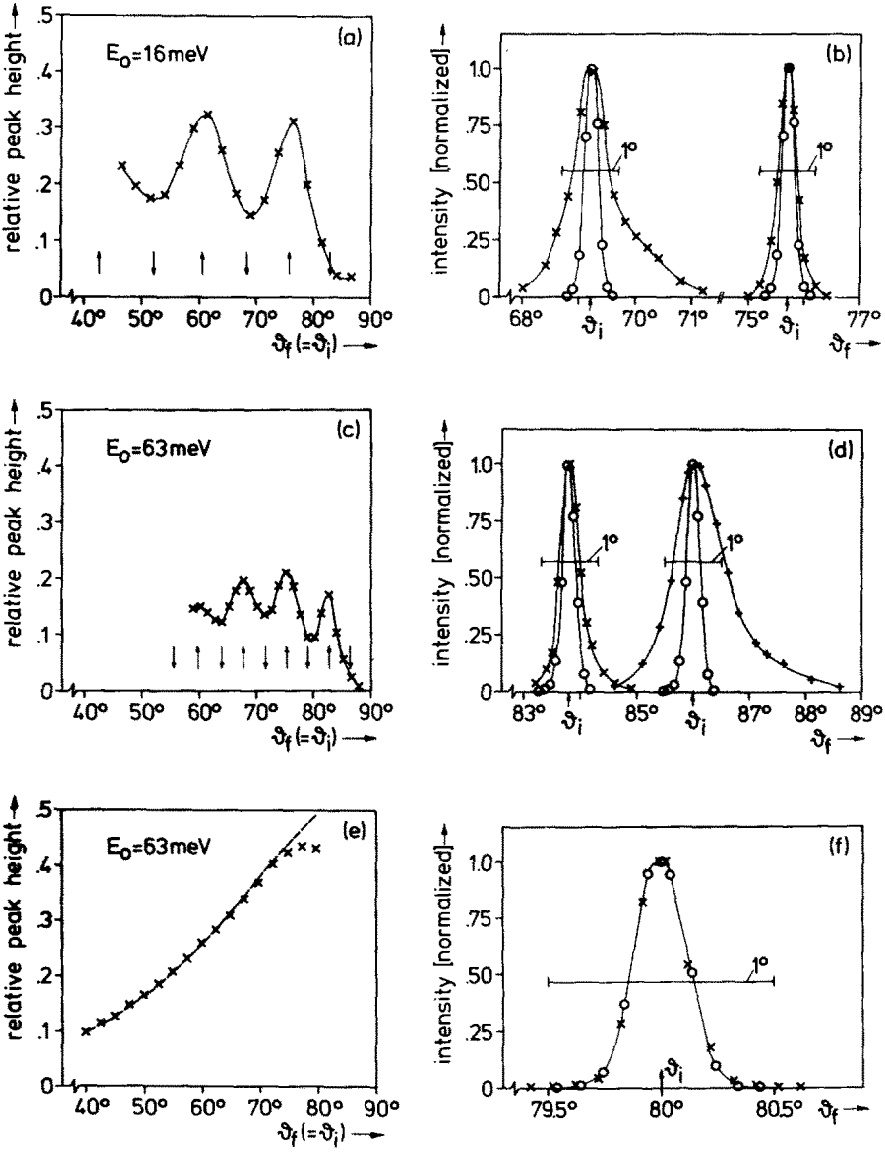


Fig. 1. Specular peak height angular distributions relative to the direct beam on the left, and normalized incident (○) and reflected (×) beam profiles on the right. Results obtained after the initial target preparation (a–d) reveal interference effects. The positions of the maxima (↑) and minima (↓) correspond to the Pt(111) interlayer spacing of 2.27 Å. The data in (e) and (f) were obtained after re-orienting and repolishing the crystal. The data presented above were measured at a target temperature of about 450 K.

0.4–0.5° with respect to the (111) plane.

Fig. 1b gives normalized scattered beam profiles for two angles of incidence of the 16 meV helium beam. The profile for scattering at an angle of incidence of $\vartheta_i = 69.2°$, where a minimum occurs in the peak height distribution (fig. 1a), is significantly broader than the incident beam profile. This is the expected result of the destructive interference condition [3]. The profile of the scattered beam measured at $\vartheta_i = 75.8°$ is still somewhat broadened ($\sim 0.1°$) relative to the incident beam and is evidence of additional surface defects. Similar behaviour is seen for the profiles obtained with the room temperature nozzle beam (63 meV) which are presented in fig. 1d. It may be stressed that the present experimental conditions allow a high sensitivity for observing structural defects. If, for example, the acceptance angle of the detector were increased from 0.15° to 0.5°, these structural effects would then only be seen at glancing angles.

Following the measurements 1a–1d, the platinum crystal was repolished following essentially the same procedure, although greater care was taken to achieve an alignment of the polished surface to within 0.1° of the (111) plane. The surface was then cleaned and annealed as previously described whereupon the surface was found, by means of the helium scattering, to have been greatly improved. After more sputtering and heating cycles all measureable structural artifacts were eliminated from the scattering distributions. This is demonstrated by the scattering data presented in figs. 1e–1f. Note first that the specular peak-height scan is free of observable oscillations and, in fact, follows the angular dependence expected from purely Debye–Waller type scattering out to about $\vartheta_i = 75°$. The precision of these data imply an "average" terrace width of greater than 3000 Å; i.e., a misalignment of the crystal surface of less than 0.05°. The dashed curve is a calculated Debye–Waller factor variation with angle of incidence. It corresponds to a surface Debye temperature of 231 K which is close to the value of 240 K given in the literature for the bulk. However, we do not suggest that this is necessarily an accurate determination of the surface Debye temperature but is only presented as a further indication that the interactions have been reduced to the intrinsic case of Debye–Waller type scattering. This proposal is also supported by comparing the incident and reflected beam profiles give in fig. 1f, where the distributions are essentially indistinguishable. It is expected that some quasi-elastic scattering, for example single phonon exchange, does occur at angles near the specular, but this appears to be a very minor contribution and can be neglected for the purpose of using the helium specular beam as a probe of the surface.

The scattering distributions plotted as a fraction of the incident beam should not be considered accurate to better than ±20%. Nevertheless, it is clear that the peak height at the specular angle is significantly greater for the re-polished crystal, even when compared at the maxima in the oscillations *.

* Calculation of the "Debye–Waller" curve in fig. 1e was based on the experimental value of I/I_0 at $\vartheta_i = 40°$ and resulted in a value for the depth of the well (Beeby correction) of 12 meV. Using a perhaps more realistic value of 6 meV, however, shifts the ordinate only 10%.

This can be attributed only in part to the slight change (0.1° FWHM) in the scattered beam profile compared with the first preparation of the crystal. Evidently some areas of the crystal after the first preparation had additional defects that did not reflect specularly.

If one wishes to use the peak height of the specularly scattered beam as a measure of adsorbate coverage as has been done previously, one question which arises is whether the scattered beam profile changes with coverage. This issue is also of more fundamental interest as it also relates to the details of the scattering interaction. Fig. 2 gives, on an expanded scale, normalized scattered beam profiles before and after adsorption of H_2 and CO. Within the resolution of the experiment the reflected distributions, after significant reductions of the reflected intensity, remain unchanged. This result, although somewhat unexpected, does simplify the technique since the coverage can be simply related to the peak height rather than requiring integration of the total peak intensity. This same result was also observed for C_2H_2 and O_2 adsorption so the conclusion does appear to be rather general, at least in practice for helium scattering.

The final question to be addressed here concerns the dependence of the attenuation of the scattered beam on incident beam angle and energy. Here H_2 and CO present two distinct cases. In fig. 3a are given results for the attenuation of the specular peak height, I/I_0, at fixed hydrogen coverage as a function of angle of incidence for two helium beam energies, 63 and 165 meV. At 63 meV there is an essentially monotonic decrease in the adsorbate induced attenuation toward glancing incidence. The minimum in the 165 meV data suggests the possibility of interference effects due to a height difference associated with the hydrogen adsorption sites relative to the uncovered surface. Application of eq. (1) yields a height difference of 0.2 Å, a value that compares

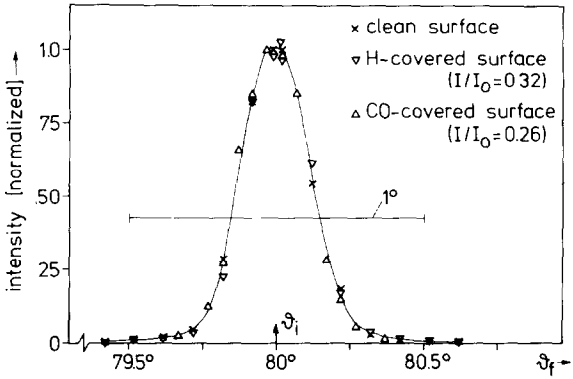


Fig. 2. Specular beam profiles ($\vartheta_i = 80°$) normalized at their peak heights in the absence (×) and presence of adsorbates. Results for H (▽) and CO (△) are shown at relative peak heights (I/I_0) of respectively 0.32 and 0.26.

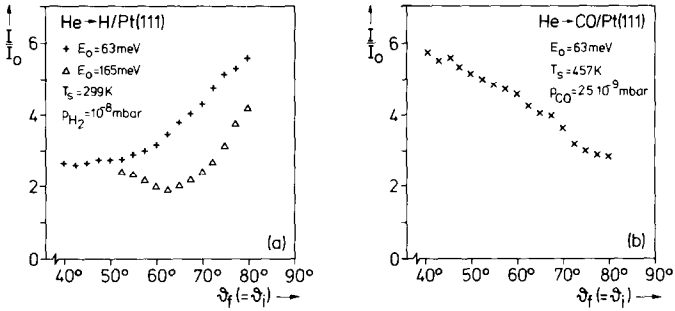


Fig. 3. Angular dependence of the specular peak height ratio (I/I_0) for the clean (I_0) and the adsorbate covered (I) surface. Results are shown for hydrogen (a) and carbon monoxide (b).

favorably with the one obtained by Engel and Rieder [5]. The corresponding minimum in the 63 meV data would be expected near $\vartheta_i = 40°$. This certainly does not exclude Debye–Waller scattering which could also qualitatively explain the increased sensitivity of the helium scattering to the adsorbed hydrogen for smaller angles of incidence and higher energies. The absence of significant beam energy and angular dependences at angles less than $\vartheta_i = 50°$ is in agreement with previous results for hydrogen adsorption on Pt(S)–[9(111)×(11$\bar{1}$)].

Fig. 3b gives comparable results for CO adsorption. In this case, the cross-section for adsorbate scattering increases (i.e., the specular intensity decreases) toward glancing incidence and there is no important effect of beam energy over the whole range of incident angles. (If the slight ripple on the curve is to be interpreted in terms of interference oscillations, it would indicate a reasonable height difference of 1.7 Å.) The increased scattering cross-section toward glancing incidence suggests that a "shadowing" effect is dominating, at least at large angles, relative to Debye–Waller scattering. Shadowing is a rather classical concept based on some assumed shape of the adsorbate on the surface which includes some lateral as well as height dimension. From such an argument it is only possible to infer some very approximate idea of the relative dimensions of the adsorbate as seen by the helium atom. Debye–Waller scattering and interference effects may have obscured relatively small shadowing effects for the case of hydrogen or, alternatively, it is possible that hydrogen induces minima in the surface potential as suggested recently by Rieder et al. [6].

A determination of the absolute cross-sections for helium scattering from adsorbates requires an independent determination of the absolute adsorbate coverage. We are currently carrying out such measurements using work function and ion scattering. For CO, it is clear that the scattering cross-section is considerably larger than the Van der Waals diameter of the molecule. Moreover, the apparent cross-section increases with the degree of perfection of the

surface. The results for hydrogen suggest, although less persuasively, a similar trend. These results have important implications and will be presented in more detail in a subsequent report.

4. Conclusion

There are several advantages to using thermal energy atom scattering for studying surfaces. These include: (1) exclusive surface sensitivity, (2) non-perturbing, and (3) the ability to abstract surface potential structure information. However, the fruitful application of the technique obviously requires some development as have other techniques such as LEED, AES, ESCA, etc. This study has demonstrated certain capabilities of the technique, and that which is presently best understood is the interpretation of the scattering relative to the structure of the surface itself. In fact, the large structure effects due to what is normally considered a "good" surface; i.e. with ~ 300 Å terraces, gave an impetus to improvements in the surface preparation technique resulting in a surface with terraces typically larger than 3000 Å. We have evidence that differences in structure density of this order quantitatively affect the behaviour of adsorbates such a H and CO. In particular, for CO the very large helium scattering cross-section compared with the nominal Van der Waals diameter suggests a long-range interaction with the surface, perhaps by means of a dipole interaction as suggested recently by Moskovits and Hulse [7].

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